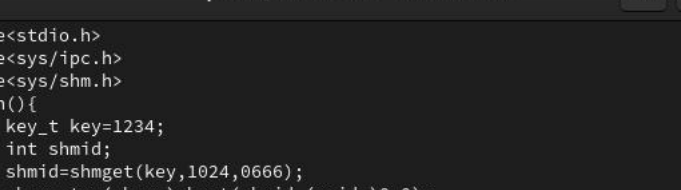


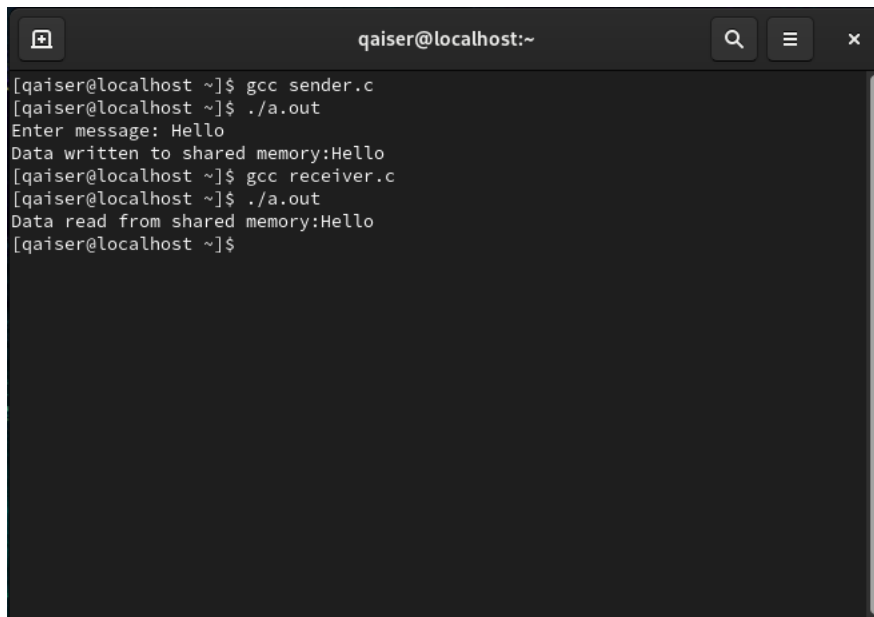
Ex. 7 Inter Process Communication Using Semaphores

```
qaiser@localhost:~ — vi sender.c
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<string.h>
int main(){
    key_t key=1234;
    int shmid;
    shmid=shmget(key,1024,0666|IPC_CREAT);
    char*str=(char*)shmat(shmid,(void*)0,0);
    printf("Enter message: ");
    scanf("%s", str);
    printf("Data written to shared memory:%s\n",str);
    return 0;
}

-- INSERT --
```



```
qaiser@localhost:~ — vi receiver.c
#include<stdio.h>
#include<sys/ipc.h>
#include<sys/shm.h>
int main(){
    key_t key=1234;
    int shmid;
    shmid=shmget(key,1024,0666);
    char*str=(char*)shmat(shmid,(void*)0,0);
    printf("Data read from shared memory:%s\n",str);
    shmdt(str);
    return 0;
}
```



```
qaiser@localhost:~  
[qaiser@localhost ~]$ gcc sender.c  
[qaiser@localhost ~]$ ./a.out  
Enter message: Hello  
Data written to shared memory:Hello  
[qaiser@localhost ~]$ gcc receiver.c  
[qaiser@localhost ~]$ ./a.out  
Data read from shared memory:Hello  
[qaiser@localhost ~]$
```

The image shows a terminal window with a dark background. The title bar at the top reads "qaiser@localhost:~". On the right side of the title bar are three icons: a magnifying glass, a hamburger menu, and a close button. The terminal content shows a series of commands and their outputs. First, the user runs "gcc sender.c", then "./a.out". They are prompted to "Enter message:" and type "Hello". The output is "Data written to shared memory:Hello". Next, they run "gcc receiver.c", then "./a.out", and the output is "Data read from shared memory:Hello". The prompt "[qaiser@localhost ~]" appears at the end of each line.